MH3600 Topology and Manifolds

Solutions of Tutorial #6

Topics covered:

- Topologies on product spaces and continuity
- More about metric spaces

Problem 1

Consider the product spaces $\prod X_\alpha$ and $\prod Y_\alpha$ in the box topologies over the index set J. Prove that if $f : \prod X_\alpha \to \prod Y_\alpha$ defined by

$$f\left((x_\alpha)_{\alpha\in J}\right) = \left(f_\alpha\left(x_\alpha\right)\right)_{\alpha\in J}$$

and each $f_\alpha : X_\alpha \to Y_\alpha$ is continuous, then f is continuous. Additionally:

(i) Does this conclusion hold true when $\prod X_\alpha$ and $\prod Y_\alpha$ are endowed with the product topologies? Justify your answer.

(ii) Does this conclusion hold true when $\prod X_\alpha$ is endowed with the box topology and and $\prod Y_\alpha$ is endowed with the product topology? Justify your answer.

(iii) Does this conclusion hold true when $\prod X_\alpha$ is endowed with the product topology and and $\prod Y_\alpha$ is endowed with the box topology? Justify your answer.

Solution: Consider any basis element $B = \prod V_\alpha$ in $\prod Y_\alpha$ so that each V_α is open in Y_α since we are in the box topology. For each $\alpha \in J$ then define $U_\alpha = f_\alpha^{-1}\left(V_\alpha\right)$, which is open in X_α since f_α is continuous. Hence the set $U = \prod U_\alpha$ is a basis element of $\prod X_\alpha$ in the box topology and is therefore open. We claim that $U = f^{-1}(B)$, which shows that f is continuous since U is open and B was arbitrary.

($\subseteq$) Consider any $\mathbf{x} \in U = \prod U_\alpha$. Then, for any $\alpha \in J$, we have $x_\alpha \in U_\alpha = f_\alpha^{-1}\left(V_\alpha\right)$ so that $f\left(x_\alpha\right) \in V_\alpha$. Hence $f(\mathbf{x}) = \left(f_\alpha\left(x_\alpha\right)\right)_{\alpha\in J} \in \Pi V_\alpha = B$ so that $\mathbf{x} \in f^{-1}(B)$. this shows that $U \subset f^{-1}(B)$ since $\mathbf{x}$ was arbitrary.

($\supseteq$) Now consider any $\mathbf{x} \in f^{-1}(B)$ so that $f(\mathbf{x}) \in B = \prod V_\alpha$ and hence each $f_\alpha\left(x_\alpha\right) \in V_\alpha$ by the definition of f. Then $x_\alpha \in f_\alpha^{-1}\left(V_\alpha\right) = U_\alpha$ so that clearly $\mathbf{x} \in \prod U_\alpha = U$. Since $\mathbf{x}$ was arbitrary this shows that $f^{-1}(B) \subset U$ as well.

We now answer the remaining three points:

(i) Yes, f is continuous because it can be equivalently rewritten as

$$f\left((x_\alpha)_{\alpha\in J}\right) = \left(\overline{f_\beta}\left((x_\alpha)_{\alpha\in J}\right)\right)_{\beta\in J} \tag{1}$$

where $\overline{f_\beta} : \prod X_\alpha \to Y_\beta$ is defined by $\overline{f_\beta} = f_\beta \circ \pi_\beta$ and $\pi_\beta : \prod X_\alpha \to Y_\beta$ is the projection $\pi_\beta\left((x_\alpha)_{\alpha\in J}\right) = x_\beta$, all of this for any $\beta \in J$. Given that π_β is continuous when $\prod X_\alpha$ is endowed with the product topology, this implies that so is the composition $\overline{f_\beta} = f_\beta \circ \pi_\beta$ and hence by (1) we have written f as a product of continuous functions that, by Theorem 19.6, is continuous.

(ii) It is also true. This conclusion can be obtained my mimicking the previous proof and by noting the π_β is also continuous when $\prod X_\alpha$ is endowed with the box topology.

(iii) This is not true. As a counterexample suppose that $X_\alpha = Y_\alpha$ for each $\alpha \in J$ and assume that f is the identity map, where the domain is endowed with the product topology and the codomain with the box topology.

Problem 2

Given sequences $(x_1, x_2, \ldots)$ and $(b_1, b_2, \ldots)$ of reals numbers with $a_i > 0$ for all $i \in \mathbb{N}$, define $h : \mathbb{R}^\omega \to \mathbb{R}^\omega$ by the equation

$$h\left((x_1, x_2, \ldots)\right) = \left(a_1 x_1 + b_1, a_2 x_2 + b_2, \ldots\right).$$

Show that if $\mathbb{R}^\omega$ is given the product topology, h is a homeomorphism of $\mathbb{R}^\omega$ with itself. What happens if $\mathbb{R}^\omega$ is given the box topology?

Solution: First note that clearly $h(\mathbf{x}) = (h_1(\mathbf{x}), h_2(\mathbf{x}), \ldots)$ for $\mathbf{x} \in \mathbb{R}^\omega$, where each $h_i : \mathbb{R}^\omega \to \mathbb{R}$ is defined by

$$h_i(\mathbf{x}) = a_i \pi_i(\mathbf{x}) + b_i$$

This can further be broken down as $h_i(\mathbf{x}) = f_i\left(\pi_i(\mathbf{x})\right) = \left(f_i \circ \pi_i\right)(\mathbf{x})$, where each $f_i : \mathbb{R} \to \mathbb{R}$ is defined by $f_i(x) = a_i x + b_i$. As discussed in the proof of Theorem 19.6, each π_i is continuous and we have that each f_i is continuous by elementary calculus, noting that this is true whether each $a_i > 0$ or not. It then follows from Theorem 18.2 part (c) that each $f_i \circ \pi_i = h_i$ is continuous. Then we have that h is continuous by Theorem 19.6 since each coordinate function is continuous and we are using the product topology.

Now define the functions $g_i : \mathbb{R} \to \mathbb{R}$ by $g_i(x) = (x - b_i)/a_i$ for $i \in \mathbb{Z}_+$, noting that this is defined since each $a_i > 0$. Define also the functions $k_i : \mathbb{R}^\omega \to \mathbb{R}$ by $k_i = g_i \circ \pi_i$, and finally define $k : \mathbb{R}^\omega \to \mathbb{R}^\omega$ by $k(\mathbf{x}) = (k_1(\mathbf{x}), k_2(\mathbf{x}), \ldots)$. Now again we have that each π_i and g_i are continuous by the proof of Theorem 19.6 and elementary calculus. Hence $k_i = g_i \circ \pi_i$ and k are continuous by Theorem 18.2 part (c), and Theorem 19.6, respectively, as before.

Now consider any $\mathbf{x} = (x_1, x_2, \ldots) \in \mathbb{R}^\omega$ so that we have, for any $i \in \mathbb{Z}_+$,

$$\begin{aligned} k_i(h(\mathbf{x})) &= [g_i \circ \pi_i] (h(\mathbf{x})) = g_i (\pi_i(h(\mathbf{x}))) = g_i (h_i(\mathbf{x})) \\ &= g_i ([f_i \circ \pi_i] (\mathbf{x})) = g_i (f_i (\pi_i(\mathbf{x}))) = g_i (f_i (x_i)) \\ &= \frac{f_i (x_i) - b_i}{a_i} = \frac{(a_i x_i + b_i) - b_i}{a_i} = \frac{a_i x_i}{a_i} \\ &= x_i \end{aligned}$$

Therefore

$$k(h(\mathbf{x})) = (k_1(h(\mathbf{x})), k_2(h(\mathbf{x})), \ldots) = (x_1, x_2, \ldots) = \mathbf{x}$$

We also have that

$$\begin{aligned} h_i(k(\mathbf{x})) &= [f_i \circ \pi_i] (k(\mathbf{x})) = f_i (\pi_i(k(\mathbf{x}))) = f_i (k_i(\mathbf{x})) \\ &= f_i ([g_i \circ \pi_i] (\mathbf{x})) = f_i (g_i (\pi_i(\mathbf{x}))) = f_i (g_i (x_i)) \\ &= a_i g_i (x_i) + b_i = a_i \left(\frac{x_i - b_i}{a_i} \right) + b_i = (x_i - b_i) + b_i \\ &= x_i \end{aligned}$$

for each $i \in \mathbb{Z}_+$so that

$$h(k(\mathbf{x})) = (h_1(k(\mathbf{x})), h_2(k(\mathbf{x})), \ldots) = (x_1, x_2, \ldots) = \mathbf{x}.$$

Since $\mathbf{x}$ was arbitrary, it thus follows from Lemma 2.1 that h is bijective and $k = h^{-1}$. Since we have already shown that h and $k = h^{-1}$ are continuous, this suffices to prove that h is a homeomorphism as desired.

We claim that h is also a homeomorphism in the box topology. Indeed, h is still a bijection as the proof of this above does not depend on the topology at all. However, Theorem 19.6 was used in the proofs that h and h^{-1} are continuous, and we know that this theorem is not generally true for the box topology. On the other hand h can be formulated as $h(\mathbf{x}) = (f_1 (x_1), f_2 (x_2), \ldots)$, where as before each $f_i(x) = a_i x + b_i$. Since each f_i is continuous by elementary calculus, it follows from Lemma 19.8.1 that h is continuous in the box topology. The same argument applies to the inverse function h^{-1} since $h^{-1}(\mathbf{x}) = (g_1 (x_1), g_2 (x_2), \ldots)$ and each g_i is continuous. ■

Problem 3

Let X be metric space with metric d.

(a) Show that $d : X \times X \to \mathbb{R}$ is continuous.

(b) Let X' denote a space having the same underlying set as X. Show that if $d : X' \times X' \to \mathbb{R}$ is continuous, then the topology of X' is finer than the topology of X.

One can summarize the result of this exercise as follows: If X has a metric d, then the topology induced by d is the coarsest topology relative to which the function d is continuous.

Solution: (a) We use Theorem 18.1 part (4) to show that d is continuous. So consider any $x_1 \times x_2 \in X \times X$ and any neighborhood V of $z = d(x_1, x_2)$, noting that $V \subset \mathbb{R}$ since $\mathbb{R}$ is the range of d. Since V is open in $\mathbb{R}$, there is a basis element $B = (a, b)$ containing z where $B \subset V$. Hence $a < z < b$. Now let $\epsilon = \min\{(z-a)/2, (b-z)/2\}$, noting that $\epsilon > 0$ since $z > a$ and $b > z$. Next define $U_1 = B_d(x_1, \epsilon)$ and $U_2 = B_d(x_2, \epsilon)$ so that they are both basis elements and therefore open sets of the metric space X. It then follows that $U = U_1 \times U_2$ is a a basis element and therefore an open set of the product space $X \times X$. Clearly we have that $x_1 \in B_d(x_1, \epsilon) = U_1$ and $x_2 \in B_d(x_2, \epsilon) = U_2$ so that U contains $x_1 \times x_2$ and so is a neighborhood of $x_1 \times x_2$.

We claim that $d(U) \subset B$. To see this, consider any $w \in d(U)$ so that there is a $y_1 \times y_2 \in U = U_1 \times U_2$ such that $w = d(y_1, y_2)$. Therefore $y_1 \in U_1 = B_d(x_1, \epsilon)$ so that $d(y_1, x_1) < \epsilon$, and similarly $d(y_2, x_2) < \epsilon$ since $y_2 \in U_2 = B_d(x_2, \epsilon)$. Then, since d is a metric, we have

$$\begin{aligned}
z = d(x_1, x_2) &\leq d(x_1, y_1) + d(y_1, x_2) \\
&\leq d(x_1, y_1) + d(y_1, y_2) + d(y_2, x_2) \\
&= d(y_1, x_1) + d(y_1, y_2) + d(y_2, x_2) \\
&< \epsilon + w + \epsilon \\
z &< w + 2\epsilon \leq w + 2(z-a)/2 = w + z - a \\
a &< w.
\end{aligned}$$

Similarly, we have

$$\begin{aligned}
w = d(y_1, y_2) &\leq d(y_1, x_1) + d(x_1, y_2) \\
&\leq d(y_1, x_1) + d(x_1, x_2) + d(x_2, y_2) \\
&= d(y_1, x_1) + d(x_1, x_2) + d(y_2, x_2) \\
&< \epsilon + z + \epsilon \\
w &< z + 2\epsilon \leq z + 2(b-z)/2 = z + b - z \\
w &< b.
\end{aligned}$$

We therefore have that $a < w < b$ so that $w \in (a, b) = B$. This of course shows that $d(U) \subset B$ since w was arbitrary. Moreover, we have that $B \subset V$ so that clearly $d(U) \subset V$, which completes the proof of Theorem 18.1 part (4) so that d is continuous.

(b) Let U be any open set of X and consider any $x \in U$. Then clearly there is a basis element $B_d(y, \epsilon)$, for some $\epsilon > 0$ and $y \in U$, of the metric topology X that contains x and where $B_d(y, \epsilon) \subset U$. Now, since d is continuous with respect to $X' \times X'$, it follows from Exercise 18.11 that the function $d_y(z) = d(y, z)$ is a continuous function from X' to $\mathbb{R}$. Since clearly the interval $(-\infty, \epsilon)$ is open in $\mathbb{R}$, it then follows that the set

$$d_y^{-1}((-\infty, \epsilon)) = \{z \in X' \mid d_y(z) < \epsilon\} = \{z \in X \mid d(y, z) < \epsilon\} = B_d(y, \epsilon)$$

is also open in X'. Thus $B_d(y, \epsilon)$ is an open set in X' containing x such that $B_d(y, \epsilon) \subset U$. This shows that U is also open in X' since the point $x \in U$ was arbitrary. This suffices to show the desired result. $\blacksquare$

Problem 4

Suppose that X is a topological space and Y and Y' are topological spaces on the same set, and that Y' is finer than Y. Suppose also that $f : X \to Y$ so that of course it is also a function from X to Y'. Prove the following statements:

(1) If f is continuous with respect to Y' then it is also continuous with respect to Y.

(2) If f is not continuous with respect to Y then it is also not continuous with respect to Y'.

(3) If a sequence in Y' converges to a point y_0, then it also converges to y_0 in Y.

(4) If a sequence in Y does not converge to a point y_0, then it also does not converge to y_0 in Y'.

(5) If a sequence in Y does not converge at all, then it also does not converge at all in Y'.

(6) If Y is a Hausdorff space, then so is Y'.

Solution: For assertion **(1)** suppose that f is continuous with respect to Y' and let U be any open set of Y. Since Y' is finer than Y, it follows that U is also open in Y'. Then, since f is continuous with respect to Y' we have that $f^{-1}(U)$ is open in X, which suffices to show that f is continuous with respect to Y since U was an arbitrary open set.

Assertion **(2)** follows immediately from the contrapositive of **(1)**. Regarding **(3)**, suppose that a sequence $(y_1, y_2, \ldots)$ converges to y_0 in Y' and let U be any neighborhood of y_0 in Y. Then U is also open in Y' since it is finer than Y, hence U is a neighborhood of y_0 in Y'. Thus there is an $N \in \mathbb{Z}_+$such that $x_n \in U$ for all $n \geq N$, since the sequence converges to y_0 in Y'. Since U was an arbitrary neighborhood of Y, this shows that the sequence converges to y_0 in Y. Assertion **(4)** follows immediately from the contrapositive of **(3)**. Assertion **(5)** then immediately follows from **(4)** since, if a sequence does not converge at all in Y then for any point $y_0 \in Y$, it does not converge to y_0. Then it also does not converge to y_0 in Y' by **(4)**. Since y_0 was arbitrary, this shows that it does not converge at all in Y'.

For **(6)**, suppose that Y is a Hausdorff space and let x and y be distinct points of Y' so that they are of course also points of Y. Hence there are neighborhoods U and V of x and y, respectively, in Y that are disjoint. Since Y' is finer than Y, we have that U and V are also open sets of Y' and thus are disjoint neighborhoods of x and y in Y' as well. This suffices to show that Y' is Hausdorff as desired.

Problem 5

Show that if d is a metric for X, then

$$d'(x,y) = d(x,y)/(1+d(x,y))$$

is a bounded metric that gives the same topology of X.

Hint: If $f(x) = x/(1+x)$ for $x > 0$, use the mean value theorem to show that $f(a+b) - f(b) \leq f(a)$.

Solution: First we show that d' is a valid metric. Since d is a metric, we have that

$$\begin{aligned} d(x,y) &\geq 0 \\ 1 + d(x,y) &\geq 1 > 0. \end{aligned}$$

Hence clearly $d'(x,y) = d(x,y)/(1+d(x,y)) \geq 0$ as well. If $x = y$ then $d(x,y) = 0$ so that

$$d'(x,y) = \frac{d(x,y)}{1+d(x,y)} = \frac{0}{1+0} = 0.$$

Conversely, if $d'(x,y) = 0$ then it has to be that $d(x,y) = 0$ as well since $1 + d(x,y) > 0$. It then must be that $x = y$ since d is a metric. This shows the first condition for the definition of a metric.
Symmetricity is easy to show:

$$d'(x,y) = \frac{d(x,y)}{1+d(x,y)} = \frac{d(y,x)}{1+d(y,x)} = d'(y,x)$$

since again d is a metric so that $d(x,y) = d(y,x)$.

Regarding the triangle inequality, define the function $f(x) = x/(1+x)$ over the domain $\{x \in \mathbb{R} \mid x \geq 0\}$ so that clearly $d'(x,y) = f(d(x,y))$. We first show that f is monotonically increasing. The easiest way to do this is to show that its derivative is always positive, from which monotonicity follows from elementary calculus. Using the quotient rule, we have

$$f'(x) = \frac{1 \cdot (1+x) - x \cdot 1}{(1+x)^2} = \frac{1+x-x}{(1+x)^2} = \frac{1}{(1+x)^2}.$$

Clearly we have $x \geq 0$ so that $1 + x \geq 1 > 0$, and hence $(1+x)^2 > 0^2 = 0$ by Lemma 20.1.1. It then follows that $f'(x) = 1/(1+x)^2 > 0$, and thus f is monotonically increasing. Now, for any $a, b \geq 0$ we clearly have

$$\begin{aligned} 1 + a &\leq 1 + a + b + b + ab + b^2 \\ 1 + a &\leq 1 \cdot (1+a+b) + b(1+a+b) \\ 1 + a &\leq (1+a+b)(1+b) \\ \frac{1}{(1+a+b)(1+b)} &\leq \frac{1}{1+a} \end{aligned}$$

since $1+a \geq 0$ and $1+a+b$ and $1+b$ are both non-negative so that their product is as well. It then follows that

$$\begin{aligned} f(a+b)-f(b) &= \frac{a+b}{1+a+b} - \frac{b}{1+b} = \frac{(a+b)(1+b)-b(1+a+b)}{(1+a+b)(1+b)} \\ &= \frac{a+ab+b+b^2-b-ab-b^2}{(1+a+b)(1+b)} = \frac{a}{(1+a+b)(1+b)} \\ &\leq \frac{a}{1+a} = f(a) \end{aligned}$$

by what was just shown before since $a \geq 0$. Hence we have $f(a+b) \leq f(a)+f(b)$. Since d is a metric, we then of course have that $d(x,z) \leq d(x,y)+d(y,z)$. Moreover, $f(d(x,z)) \leq f(d(x,y)+d(y,z))$ (since f is monotonically increasing), and hence

$$f(d(x,z)) \leq f(d(x,y)+d(y,z)) \leq f(d(x,y))+f(d(y,z))$$

which implies that

$$d'(x,z) \leq d'(x,y)+d'(y,z),$$

for any $x,y,z \in X$. This completes the proof that d' is a valid metric.

Now, it is easy to see that d' is bounded by 1 :

$$\begin{gathered} 0<1 \\ d(x,y) < 1+d(x,y) \\ \frac{d(x,y)}{1+d(x,y)} < 1 \\ d'(x,y) < 1 \end{gathered}$$

so that d' is a bounded metric.

Now we must show that the metric topologies induced by d and d' are the same. First we show that the topology induced by d is finer than that induced by d' using Lemma 20.2. So consider any $x \in X$ and any $\epsilon > 0$. If $\epsilon \geq 1$ set $\delta = 999$, otherwise set $\delta = \epsilon/(1-\epsilon)$ noting that in this case $\epsilon < 1$ so that $0 < 1-\epsilon$ and hence $\delta > 0$. Now consider any $y \in B_d(x,\delta)$ so that $d(y,x) < \delta$. If $\epsilon \geq 1$ then of course $d'(y,x) < 1 \leq \epsilon$ since we have previously shown the d' is bounded by 1 . On the other hand, if $\epsilon < 1$, then we have

$$\begin{gathered} d(y,x) < \delta = \frac{\epsilon}{1-\epsilon} \\ d(y,x)(1-\epsilon) < \epsilon \\ d(y,x) - \epsilon d(y,x) < \epsilon \\ d(y,x) < \epsilon + \epsilon d(y,x) = \epsilon(1+d(y,x)) \\ \frac{d(y,x)}{1+d(y,x)} < \epsilon \\ d'(x,y) < \epsilon. \end{gathered}$$

Thus either was we have $d'(y,x) < \epsilon$ so that $y \in B_{d'}(x,\epsilon)$, which of course shows that $B_d(x,\delta) \subset B_{d'}(x,\epsilon)$ since y was arbitrary. Since ϵ was arbitrary this shows the desired result that the topology induced by d is finer than that induced by d'. Now we show the other direction, i.e. that the d' topology is also finer than the d topology, again using Lemma 20.2. So consider any $x \in X$ and $\epsilon > 0$ again. This time set $\delta = \epsilon/(1+\epsilon)$ noting that $\delta > 0$ follows trivially from the fact that $\epsilon > 0$. Then, for any $y \in B_{d'}(x,\delta)$, we have that

$$\begin{aligned}
d'(y,x) < \delta &= \frac{\epsilon}{1+\epsilon} \\
\frac{d(y,x)}{1+d(y,x)} &< \frac{\epsilon}{1+\epsilon} \\
d(y,x)(1+\epsilon) &< \epsilon(1+d(y,x)) \\
d(y,x) + \epsilon d(y,x) &< \epsilon + \epsilon d(y,x) \\
d(y,x) &< \epsilon
\end{aligned}$$

so that $y \in B_d(x,\epsilon)$. This of course shows that $B_{d'}(x,\delta) \subset B_d(x,\epsilon)$, which in turn shows that the d' topology is finer than the d topology as desired. Since each topology is finer than the other, of course they must be the same as desired. ■

Problem 6

Let X and Y be metric spaces with metrics d_X and d_Y, respectively. Let $f : X \to Y$ have the property that for every pair of points x_1, x_2 of X,

$$d_Y\left(f\left(x_1\right), f\left(x_2\right)\right) = d_X\left(x_1, x_2\right).$$

Show that f is an imbedding. It is called an isometric imbedding of X in Y.

Solution: First, let $Z = f(X)$ and let f' denote the function f with the range restricted to Z so that clearly f' is surjective. First we must show that f and therefore also f' is injective, from which it clearly follows that f' is a bijection. So suppose that $x_1, x_2 \in X$ where $f\left(x_1\right) = f\left(x_2\right)$. Then we have that

$$d_X\left(x_1, x_2\right) = d_Y\left(f\left(x_1\right), f\left(x_2\right)\right) = d_Y\left(f\left(x_1\right), f\left(x_1\right)\right) = 0$$

by property (1) of the metric d_Y, so that it must be that $x_1 = x_2$ since d_X is a metric and $d_X\left(x_1, x_2\right) = 0$. This of course means that f and f' are injective and hence f' is bijective. We show that f' and f'^{-1} are both continuous using Theorem 21.1 since X and Z are both metric spaces, noting that $Z \subset Y$ is a metric space with metric $d_Z = d_Y \mid Z \times Z$ by the previous exercise. So consider any $x \in X$ and any $\epsilon > 0$, and let $\delta = \epsilon$. Then for any $x_1, x_2 \in X$ where $d_X\left(x_1, x_2\right) < \delta$ we have

$$d_Z\left(f'\left(x_1\right), f'\left(x_2\right)\right) = d_Y\left(f\left(x_1\right), f\left(x_2\right)\right) = d_X\left(x_1, x_2\right) < \delta = \epsilon,$$

which suffices to show that f' is continuous. A similar argument shows that $f'^{-1} : Z \to X$ is continuous. For $y_1, y_2 \in Z$ and $\epsilon > 0$, again let $\delta = \epsilon$ and suppose that $d_Z(y_1, y_2) < \delta$. Let $x_1 = f'^{-1}(y_1)$ and $x_2 = f'^{-1}(y_2)$. Then we have

$$\begin{aligned} d_X\left(f'^{-1}(y_1), f'^{-1}(y_2)\right) &= d_X(x_1, x_2) = d_Y(f(x_1), f(x_2)) = d_Y(f'(x_1), f'(x_2)) \\ &= d_Y\left(f'\left(f'^{-1}(y_1)\right), f'\left(f'^{-1}(y_2)\right)\right) = d_Y(y_1, y_2) \\ &= d_Z(y_1, y_2) < \delta = \epsilon \end{aligned}$$

which of course shows that f'^{-1} is also continuous. This shows that f' is a homeomorphism so that f is an imbedding as desired. ■